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Digital Coating Inspection Portal - AI Driven

Partho Mullick and Maryam Alsheeb, Central Engineering Division, ADNOC OFFSHORE, Abu Dhabi, United Arab Emirates

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Abstract

This abstract details the deployment of a sophisticated, AI-powered inspection solution built upon ADNOC Offshores existing TrueView 3D-laser-scan repository and "Virtual Tour" framework for three-dimensional model visualization. The initiative's primary aim is to generate a highly precise, end-to-end inventory of all observable external corrosion anomalies by applying advanced machine-learning techniques to the point-cloud data. By automating the detection, classification, and quantification of paint and substrate degradation across every offshore asset, this approach dramatically curtails reliance on manual, in-field visual surveys—enabling teams to allocate inspection resources more efficiently, improve consistency in condition assessments, and ultimately accelerate maintenance decision cycles while maintaining rigorous safety and reliability standards.

At present, certified inspection technicians conduct on-site visual assessments for external corrosion, which can result in incomplete coverage and introduce the potential for human error. The proposed Digital AI platform—a cloud-based web application within the Digitalization program—leverages Artificial Intelligence and Machine Learning to detect coating damage and generate a comprehensive digital record. This solution enhances inspection processes and minimizes oversight of unmanned assets, reduces personnel exposure at sites, and mitigates the risk of safety incidents.

Deployment of Digital Coating Inspection Portal will facilitate:

- Deliver a comprehensive digital solution for the assessment of external corrosion and the management of coating maintenance.
- 95% reduction in risk of failures caused by coating degradation and external corrosion
- Up to 40% reduction in Barge time through enhanced coverage and Targeted Painting
- Reduce painting & coating costs through a continuous Digitization Process
- Data-driven decisions independently or collaboratively
- Reduced site visit requirements enabling Logistic cost savings (Chopper/Boats etc.) achieve the goal of "Decarbonization"

Introduction

We rank among the world's leading offshore hydrocarbon producers, overseeing the development, production, and delivery of oil and gas resources. Since 1962, we have supplied oil for the UAE, offering products such as crude oil, raw gas, and condensate across nine established fields comprising eight major complexes and six artificial islands.

Company's portfolio includes management of Upper Zakum, Lower Zakum, Umm Shaif, Satah, Nasr, Umm Al Dalkh, Sarb, Umm Lulu, and Abu Al Bukhoosh. Operational activities further extend to our terminals on Das Island and Zirku Island, where the Company undertakes the processing, storage, and export of crude oil.

The recently developed super complexes at Nasr and Umm Lulu are advanced digital smart fields integrating state-of-the-art technology and industry best practices to promote safety, operational efficiency, and profitability. Our adoption of innovative solutions, including data integration, artificial intelligence, and autonomous robotics, continually optimizes operations and reinforces our standing as one of the industry's most cost-effective and low-carbon energy producers.

Currently, ADNOC Offshore employs traditional methods for inspecting and maintaining their assets. The traditional approach involves several direct and indirect costs associated with lack of digital data, asset maintenance issues, inspection errors, and unexpected shutdowns due to unanticipated failures.

Challenges

ADNOC Offshore is one of the largest offshore hydrocarbon producers in the world, responsible for the development, production, and delivery of oil and gas resources. All our assets are offshore based and are aging. The Structural & Piping Integrity are mostly affected because of its exposure to sea. There is a challenge to measure the severity of the Paint condition and the metal loss at nearly 300 offshore sites spread over Abu Dhabi waters. Currently only Visual inspections for external corrosion monitoring for all the assets are conducted by certified inspection technicians with physical site visits. These inspections lead to incomplete coverage, variations, and errors in reporting due to Omissions and human error in identifying, classifying, and locating external corrosion threats. Inspection and Maintenance of Offshore assets is not digital, labor intensive, has a lot of inaccuracies and subjectivity, limited coverage, difficult to communicate and has a significant carbon footprint due to logistics. This poses significant risks, leading to operational downtimes which include leakages, safety hazards, and financial losses.

The key challenges include:

- Digital Record or Database: The digital record or database serves as a crucial and permanent repository, meticulously designed to store comprehensive visual references pertinent to coating surveys and inspections. These records ensure that every inspection and survey is thoroughly documented, providing an accurate and detailed visual history. This system significantly enhances the reliability and accessibility of data, enabling precise tracking and assessment of any coating defects. By limiting these defects within the company's operations, the database supports quality control and continuous improvement processes, ensuring high standards are consistently met and maintained.
- Human Error and Subjectivity: Human error and subjectivity, often stemming from fatigue and distractions during manual inspections, can lead to overlooked defects or inaccurate assessments. When critical areas are missed, the likelihood of failure in essential business components increases dramatically. This inconsistency in inspections and maintenance planning can result in severe consequences, including costly downtime and unexpected failures. The lack of repeatability and consistency highlights the inherent flaws in relying solely on human intervention for inspections, emphasizing the need for automated solutions or enhanced training and protocols to mitigate these risks.

- Digital Maintenance Planning: Digital maintenance planning involves the complex task of accurately estimating the requirements for coating maintenance and painting at each facility. This process is critical for ensuring that all surfaces are adequately protected and maintained, which in turn affects the overall longevity and safety of the infrastructure. The challenge lies in accounting for various factors such as environmental conditions, the specific materials used, and the varying degrees of wear and tear across different areas of a facility. Advanced digital tools and predictive analytics are employed to create more precise estimates, helping to optimize resource allocation, reduce waste, and ensure timely maintenance activities. By leveraging technology, facility managers can make more informed decisions that enhance the efficiency and effectiveness of their maintenance strategies.
- Delayed Inspections: Manual inspection methods, while traditional, often prove to be inefficient as they are time-consuming and lead to significant delays in identifying and addressing issues. These delays can hinder the implementation of timely corrective actions, exacerbating the potential risks and operational inefficiencies within the plant. Furthermore, the inherent challenges of inspecting certain areas due to limited accessibility can complicate logistics, making it difficult to conduct thorough evaluations. These inaccessible areas may remain unchecked for extended periods, potentially allowing minor problems to escalate into major failures. Thus, relying solely on manual methods can compromise the overall safety and efficiency of plant operations.
- High Operational Costs: High operational costs are a significant concern for many businesses, particularly when it comes to mandatory inspections. These inspections, while crucial for ensuring safety and compliance, often lead to unavoidable downtime. During these periods, production lines come to a halt, and the temporary cessation of operations can cause a ripple effect throughout the entire supply chain. This interruption not only delays output but also requires the allocation of additional resources to manage and mitigate the impact, thereby driving up expenses. The overall financial burden is further compounded by potential penalties for non-compliance, overtime costs to catch up on lost production, and the maintenance of idle machinery. Consequently, the need for thorough and frequent inspections, though necessary, can strain a company's financial health, highlighting the ongoing challenge of balancing operational efficiency with regulatory adherence.
- Safety Risks (HSE Exposure): Manual inspections in hazardous areas present significant safety risks for personnel, as they are often exposed to dangerous conditions that can include exposure to toxic chemicals, extreme temperatures, and unstable structures. These environments can lead to serious injuries or health issues, such as chemical burns, heat strokes, or even life-threatening accidents. By placing individuals in direct contact with these hazards, the likelihood of accidents and harm increases substantially. Additionally, the physical and psychological stress of operating in such perilous settings can impair judgment and reaction time, further elevating the risk of incidents. Implementing automated or remote inspection and connected worker technologies (proposed), can mitigate these risks, safeguard personnel and ensure a safer working environment.
- Time-Consuming Inspections: Manual inspections, by their very nature, are often time-consuming and can lead to significant delays in identifying and addressing issues. This protracted process hinders timely decision-making and slows down the implementation of corrective actions, potentially exacerbating existing problems. Moreover, the effectiveness of traditional inspection methods hinges on the necessity of conducting regular follow-ups to all coating condition reports. These follow-ups are crucial to fully leverage the benefits of manual inspections, ensuring that any emerging issues are promptly identified and remedied, thereby maintaining the integrity and functionality of the inspected objects.
- Reactive vs Proactive: A reactive approach to coating maintenance often results in minor issues being neglected until they escalate into significant problems that demand costly and extensive repairs. This method relies on addressing issues after they have already caused substantial damage,

leading to increased downtime and resource consumption. In contrast, a proactive approach leverages advanced technologies such as digital databases to monitor and record the state of coatings. By providing visual references and real-time data, this method enables early detection and intervention, preventing minor issues from becoming major complications. Consequently, proactive maintenance not only minimizes costs but also enhances the longevity and performance of the coatings, ensuring a more efficient and sustainable management strategy.

- **Limited Communication of Human Reports:** One of the significant challenges in managing human reports lies in the limited communication and difficulty in sharing coating condition data among multiple stakeholders. This can hinder the efforts to optimize budgets and plan effectively. In contrast, AI-generated data is not only explainable but also fully digital, facilitating seamless sharing across all business unit levels. This digital approach ensures that all relevant parties have access to the same accurate and up-to-date information, enabling more informed decision-making and streamlined processes. By leveraging the capabilities of AI, businesses can overcome the traditional barriers of data communication and enhance overall operational efficiency.
- **Environmental Impact:** Traditional inspection methods often fail to make efficient use of resources, leading to significant waste and inefficiency. The reliance on manual inspections involves considerable carbon footprints, largely due to transportation needs, unpredictable weather considerations and prolonged procedures due to variety of operational reasons at sites. This inefficiency is further exacerbated by hydrocarbon losses stemming from failures related to coating degradation. As these manual methods persist, the environmental impact becomes substantial, contributing to increased carbon emissions and exacerbating climate change. Therefore, it is imperative to explore more sustainable and efficient inspection methods to mitigate these adverse environmental effects.

AI Driven Inspection Approach

This enterprise-grade, cloud-native platform represents a state-of-the-art solution that leverages sophisticated artificial-intelligence and machine-learning frameworks to dramatically improve the efficiency, consistency, and precision of asset inspection workflows. By shifting from manual or semi-automated methods to a fully AI-driven inspection paradigm, ADNOC Offshore can realize significant operational cost savings, compress survey timelines, and better allocate technical and field resources across its offshore installations. The digital coating management system not only accelerates coating survey execution and report generation, but also underpins an optimized, data-centric maintenance strategy covering every asset in the fleet. All coating condition assessments—complete with detailed corrosion metrics and degradation severity scores—are delivered in an immersive, interactive 3D visualization environment, accessible via the cloud to engineering, operations, inspection, HSE, and executive teams alike. By providing a single source of objective, quantifiable inspection data, this platform eradicates the ambiguity and subjectivity that often plague traditional field reviews, ensures full traceability of all inspection activities, and empowers stakeholders to make proactive, risk-informed maintenance decisions.

Level of innovation, AI maturity, project scalability

This innovative solution leverages proprietary technology to conduct comprehensive inspections, achieving an impressive asset coverage of up to 95%, which is significantly higher than the 25-40% coverage provided by traditional methods. This advanced approach not only enhances coverage but also brings substantial reductions in both cost and risk. The AI models employed have been meticulously trained to deliver 90% accuracy across numerous offshore facilities worldwide. Annually, the application process involves the analysis of 25 to 30 facilities, with a scalable capacity to expand operations to over 100 platforms. This ensures a robust and effective implementation of AI solutions, revolutionizing the inspection process and setting new standards for efficiency and reliability in the industry.

Technology used to develop the application

The platform has been architected to scale seamlessly from handling hundreds of scans per day to processing thousands, leveraging auto-scaling cloud services and distributed compute clusters. Peak throughput tests have demonstrated sub-second model inference times even as data volumes grow, ensuring the system can absorb future increases in both scan resolution and asset count without degradation in performance.

This enterprise-grade, cloud-native platform unifies cutting-edge AI-driven computer-vision inspection pipelines with a full anomaly-management stack and strategic coating-maintenance planning tools. By harnessing proprietary, patent-protected machine-learning models—trained on millions of annotated corrosion samples—the system automatically ingests high-resolution 3D laser-scan point clouds alongside synchronized photographic imagery, then performs pixel- and voxel-level analysis to detect, categorize, and measure every instance of metal loss and coating degradation. Inspection results are seamlessly overlaid into an interactive digital-twin environment hosted on scalable cloud infrastructure, where users can navigate asset geometries in real time, flag critical defects, and drill down into severity metrics. Built-in anomaly-management modules track defect evolution over time, generate prioritized repair work orders, and integrate with enterprise CMMS and ERP systems for end-to-end traceability. Key deliverables include near-real-time digital coating inspection reports, comprehensive audit trails of all detected anomalies, and automated maintenance-scope optimization that can reduce overall lifecycle costs by a substantial margin. Together, these capabilities deliver a highly efficient, data-driven solution tailored for the rigorous demands of heavy-industry asset integrity programs.

Robust error-handling logic ensures that any failed scan ingestion or model inference is automatically retried up to three times. Persistent failures are captured in an "alert" queue for manual review, guaranteeing no data is lost and providing transparency into any operational hiccups. All system actions—including scan uploads, preprocessing steps, AI inferences, and report exports—are logged with timestamps.

Features

Each stage of the workflow—from scan ingestion through preprocessing, model inference, and report generation—is implemented as a self-contained module with clearly defined interfaces. This modular design allows individual components to be swapped or upgraded (for example, replacing the corrosion classifier with a new model) without impacting the rest of the pipeline.

It utilizes the 3D Laser Scan data as a part of ADNOC Offshores established "Virtual Tour" combined with industry-leading AI. The algorithms detect all surface corrosion and coating damage, classifying the severity of corrosion based on their visual appearance, and quantify their extent using spatial analysis from the 3D data. It uses computer vision, machine learning and 3D processing to build an objective, comprehensive and accurate inventory of all observed external corrosion findings. It provides a prioritized set of inspection findings that help ADNOC OFFSHORE's Integrity Management identify and validate parts of the facility that require attention from the Maintenance and Reliability teams. Doing this digitally on scan inspection data allows for a significant reduction in visual inspections, which are otherwise sparse, labor intensive and prone to subjective biases. The process also enables the generation of a complete baseline of the facility, which will enable anomaly tracking over time with consecutive scans. Such analysis will enable ADNOC OFFSHORE to commit to a proactive framework early on and avoid a reactive corrosion management workflow, significantly reducing risk of equipment failure and potential Loss of Primary Containment (LoPC) events as the facility begins to age.

This enhanced capability enables integrity management teams to evaluate and rank every piece of process-critical equipment in a single, comprehensive scan campaign, allowing them to develop finely tuned work scopes for both mechanical integrity assessments and fabric maintenance activities. By deeply mining the 3D laser-scan and visual imagery data, the system can slash manual visual-inspection labor requirements by as much as 95%, all while seamlessly integrating with existing Risk-Based Inspection (RBI) and Reliability-Centered Maintenance (RCM) programs to proactively mitigate corrosion-related hazards.

At the heart of AI solutions are advanced computer-vision pipelines and machine-learning frameworks engineered to process multi-modal inputs—combining high-resolution photographic capture with dense point-cloud data. These algorithms automatically identify every instance of surface corrosion and coating degradation, hierarchically classify each defect’s severity based on detailed visual cues and compute the spatial footprint of damage through precise volumetric and areal analysis of the 3D scan. Concurrently, the system extracts equipment identifiers directly from the scan geometry and cross-references them against existing asset metadata and master-data sources (including CAD models, IDMS entries, and P&ID schematics) to construct a risk-ranked inventory of components. Finally, by aligning the quantified corrosion profiles with this prioritized asset list, the application produces a unified inspection-priority report that guides maintenance teams to focus efforts where they will deliver the greatest safety, reliability, and cost-efficiency benefits.

The analytics engine of the application has been tuned on a wide range of data captured globally. This enables it to accurately, objectively and scalable identify external corrosion in terabytes of data, which would otherwise not be manually possible or feasible. The system has undergone a range of independent field tests and evaluations to measure its repeatability, robustness and accuracy. Two notable tests were conducted by BP on the MadDog and Atlantis production platforms. On MadDog, a blind test of the computer vision corrosion detection algorithms showed a 97% detection accuracy of severe corrosion and over 90% on general corrosion. On the Atlantis platform, the analytics algorithms were tested on handheld imagery to see their robustness on diverse and variable data.

The solution allows for the effective utilization of the AI output through dashboards and visualizations:

- AI-powered Corrosion & Coating Condition Highlights
- Plot plan Heatmap
- Measurement Tool
- Generic Point of Interest (POI)
- Automated Report Generation
- Integration with SAP & APM
- Fleet Health Monitoring
- Quantification of Scab Corrosion and Pit Depth Analysis
- Data Filtering and Search
- Data Export and Integration
- Customizable Dashboards

The above figures ([Figure 1](#) to [5](#)) illustrate the implementation of Digital Coating in one of the assets of ADNOC Offshore as a Proof of Concept (POC) to carry out the AI driven solution utilizing Virtual Tour scan data and 3D processing to build an objective, comprehensive and accurate inventory of all observed external corrosion detection using machine learning algorithms.

This solution provided a prioritized set of inspection findings which facilitated Integrity Management to identify and validate parts of the facility that require attention from the Maintenance and Reliability teams. The AI generated report was validated by a certified paint inspector.



Figure 1—Corrosion Detection Visual

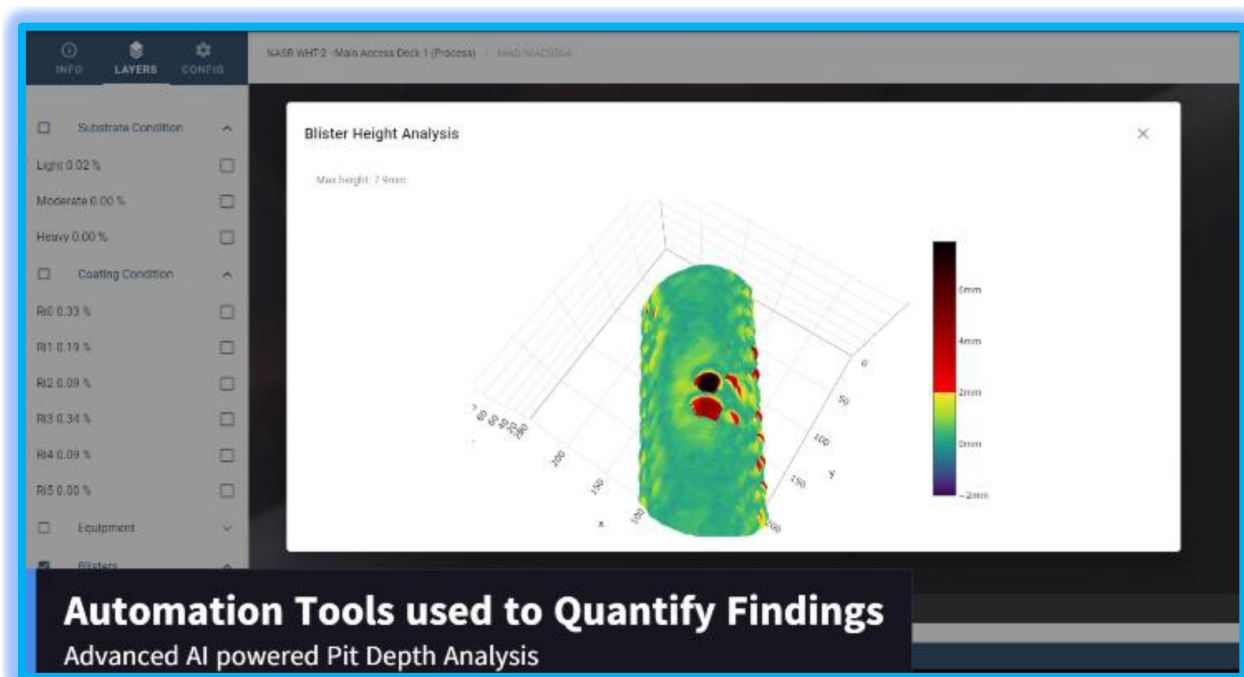


Figure 2—Blister Height Analysis – With AI



Figure 3—Component Level Package with Coating Condition

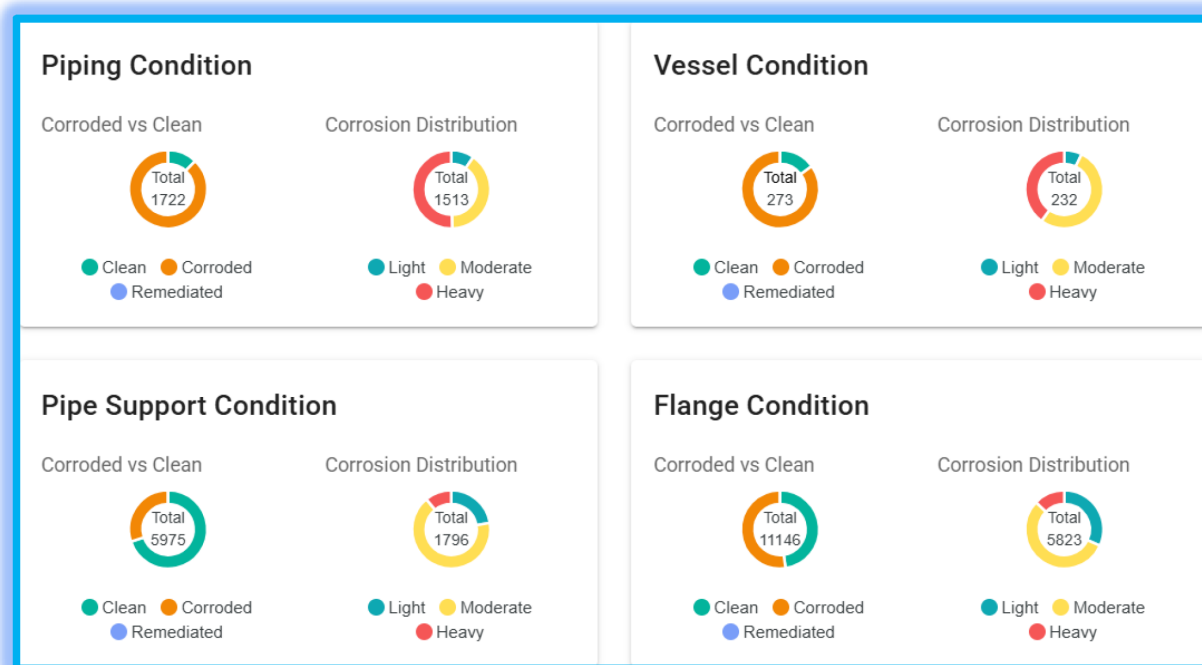


Figure 4—Dashboard illustrating the discipline wise Coating condition



Figure 5—Digital Coating Dashboard

Compliance with artificial intelligence ethics guidelines

The solution adheres to eight key AI Ethics Principles, ensuring AI is safe, secure, and reliable. These principles guide AI development to achieve fair and reliable outcomes, minimize negative impacts, and uphold the highest ethical standards. By following these principles, it ensures public trust, drives consumer loyalty, and guarantees that all users benefit from AI technology. The principles cover human and societal wellbeing, human-centered values, fairness, privacy, reliability, transparency, contestability, and accountability.

Opportunity

Impact of this idea - Implementing this technology at ADNOC Offshore will create a fully Digital Database for Offshore Inspection Management using Artificial Intelligence to automatically detect coating damage and external corrosion anomalies resulting in:

- Early Threat Detection: Proactively identifying risks to operational integrity.
- AI-Based Results of Inspection: Eliminate Subjectivity of Inspections and Add layers of consistency
- Reduced Downtime: Enabling predictive maintenance, reducing unplanned outages.
- Enhanced Safety: Lowering the risk of structural failures and personnel exposure to hazardous environments.
- Cost Savings: Reducing and planning maintenance and repair costs through early intervention.
- Environmental Benefits: Decreasing the carbon footprint through reduced site visit requirements enabling Logistic cost savings (Chopper/Boats/Barge etc.) achieving the goal of "Decarbonization".
- Comprehensive Coverage: Ensuring all critical Piping & equipment is inspected and maintained.

Value opportunity - This technology value lies in leveraging AI to enhance operational efficiency and cost-effectiveness through early anomaly detection and prioritized maintenance. This leads to:

- Extended Asset Lifespan: Preventive maintenance prolongs the life of key infrastructure which includes the leakages
- Unlock Potential of AI based Inspections: Eliminate inaccuracies and manage all offshore inspections and coating surveys digitally.
- Data Management: One integrated platform for coating inspection detection and assessment database
- Optimized Inspection Cycles: Automation reduces inspection time and human error.
- Reduced Carbon Footprint: Time efficient inspections lead to lower energy consumption and environmental impact. Reduced site visit requirements enabling Logistic cost savings (Chopper/Boats/Barge etc.) achieve the goal of "Decarbonization".
- Improved Communication: Ambiguity in classification of corrosion problems, coating defects and maintenance planning.
- Seamless Integration: This technology easily integrates with existing systems, offering a high return on investment (ROI) with quick adoption.

Results

This technology employs AI and advanced image processing technology to objectively, accurately, and comprehensively build an interactive facility corrosion & anomaly database. Additional benefits include:

Full Digital Inspection and Coating Maintenance Database (Permanent Record in visual form with high retention)

- Full Digital Inspection and Coating Maintenance Database (Permanent Record in visual form with high retention)
- Increased Coverage of Visual Inspection across all asset types (Up to ~95%)
- Eliminate human error in condition reporting using field proven AI (Reduced Subjectivity)
- Limited site exposure with high inspection clearance (Reduced Safety Hazards)
- Reduced Carbon Emissions through minimal site exposure for Maintenance (Decarbonization)
- Automated Inspections and Report (Experienced personnel can be reassigned to more critical jobs)
- Proactive Maintenance Approach (Reduced Reactive and Ad Hoc Maintenance)
- Reduce Maintenance and Paint Costs (Reduced repeat work and unnecessary remediations)
- Ethical AI application with Complete Database Access

Conclusion

The implementation of an AI-driven Digital Coating Inspection Portal marks a significant advancement in the field of corrosion management and protective coating assessment. By integrating artificial intelligence, computer vision, and data analytics, the system enables consistent, high-precision evaluation of coating conditions while minimizing reliance on manual inspection methods. This not only enhances the accuracy and repeatability of inspections but also facilitates proactive maintenance through predictive insights. The portal's ability to centralize inspection data, automate defect detection, and generate actionable reports supports improved decision-making, operational efficiency, and long-term asset integrity across industries.